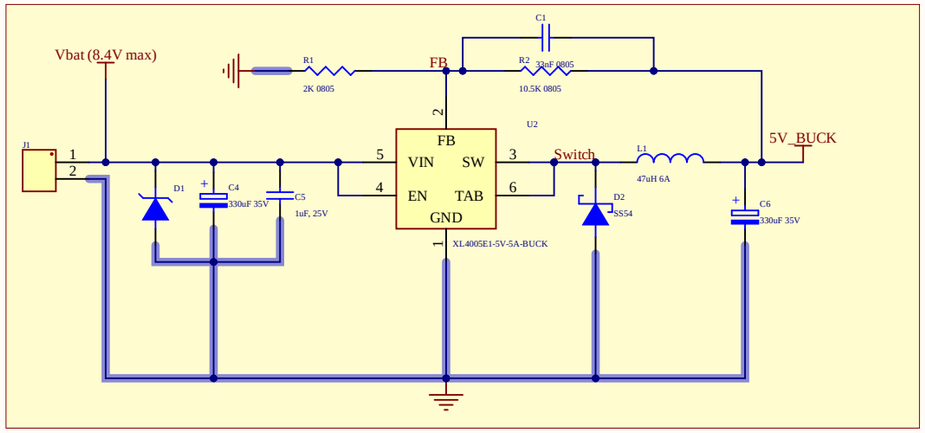
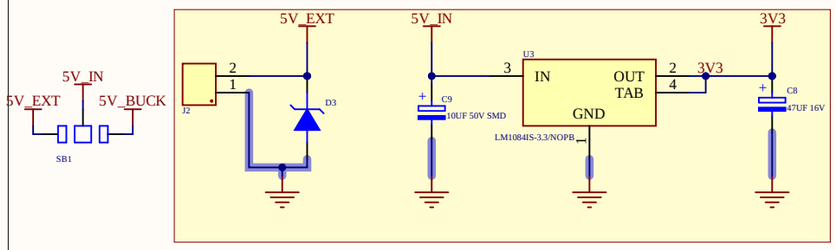
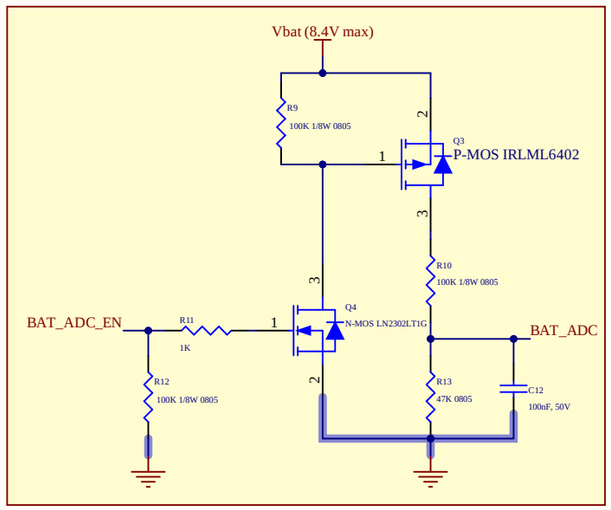


VOLTAGE REGULATOR 8V4 TO 5V

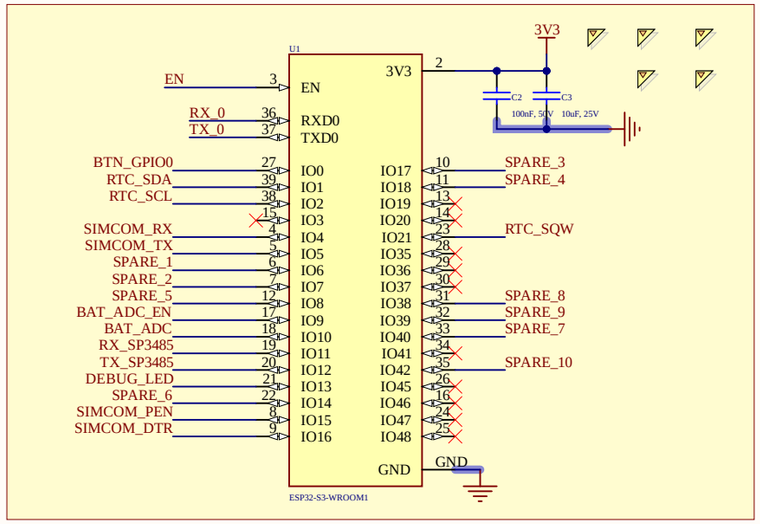


VOLTAGE REGULATOR 5V TO 3V3

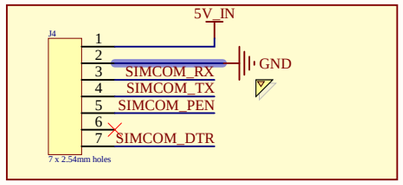
VOLTAGE OUT & MOUNTING HOLES & BULK CAPS



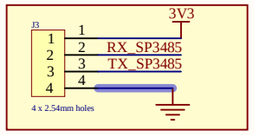
BATTERY VOLTAGE SENSING



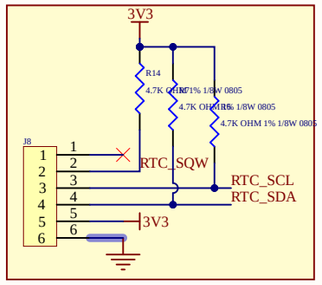
ESP32-S3 N16R8 Module



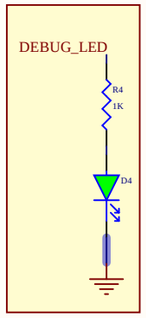
SIMCOM A7682S

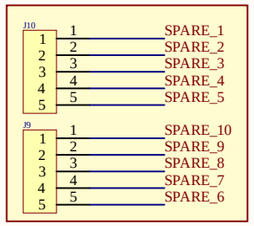


TTL to RS485 MODULE

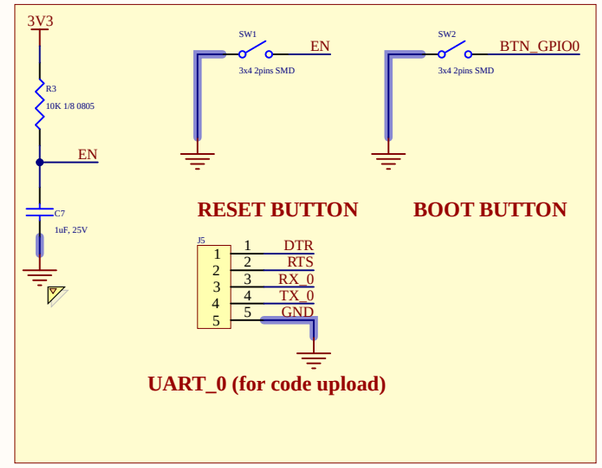


RTC MODULE DS3231 (I2C)

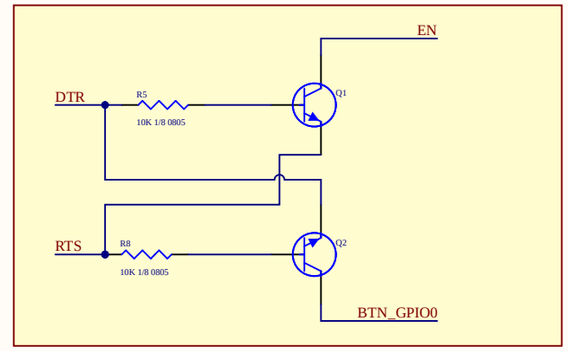




FREE PINS



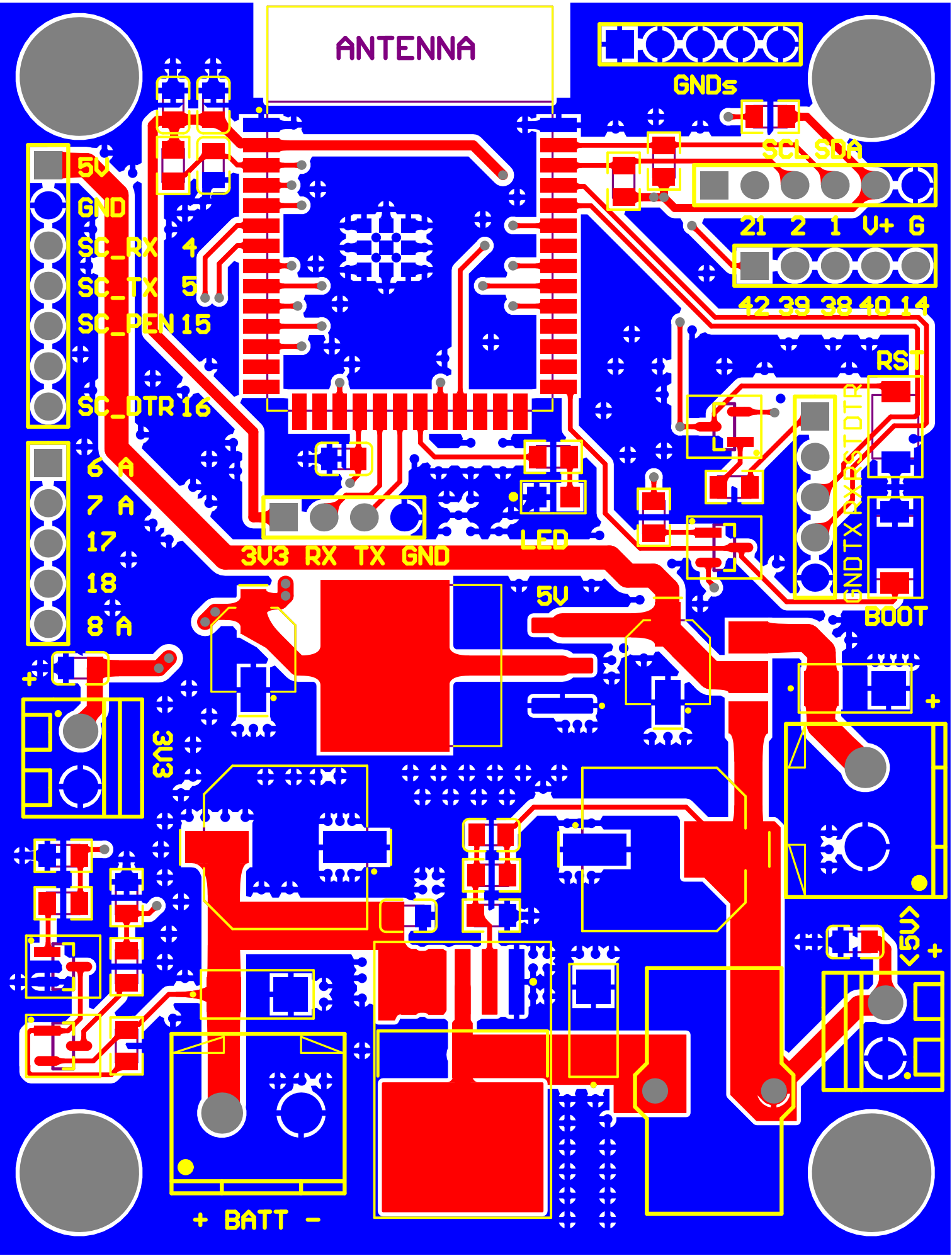
UART_0 (for code upload)

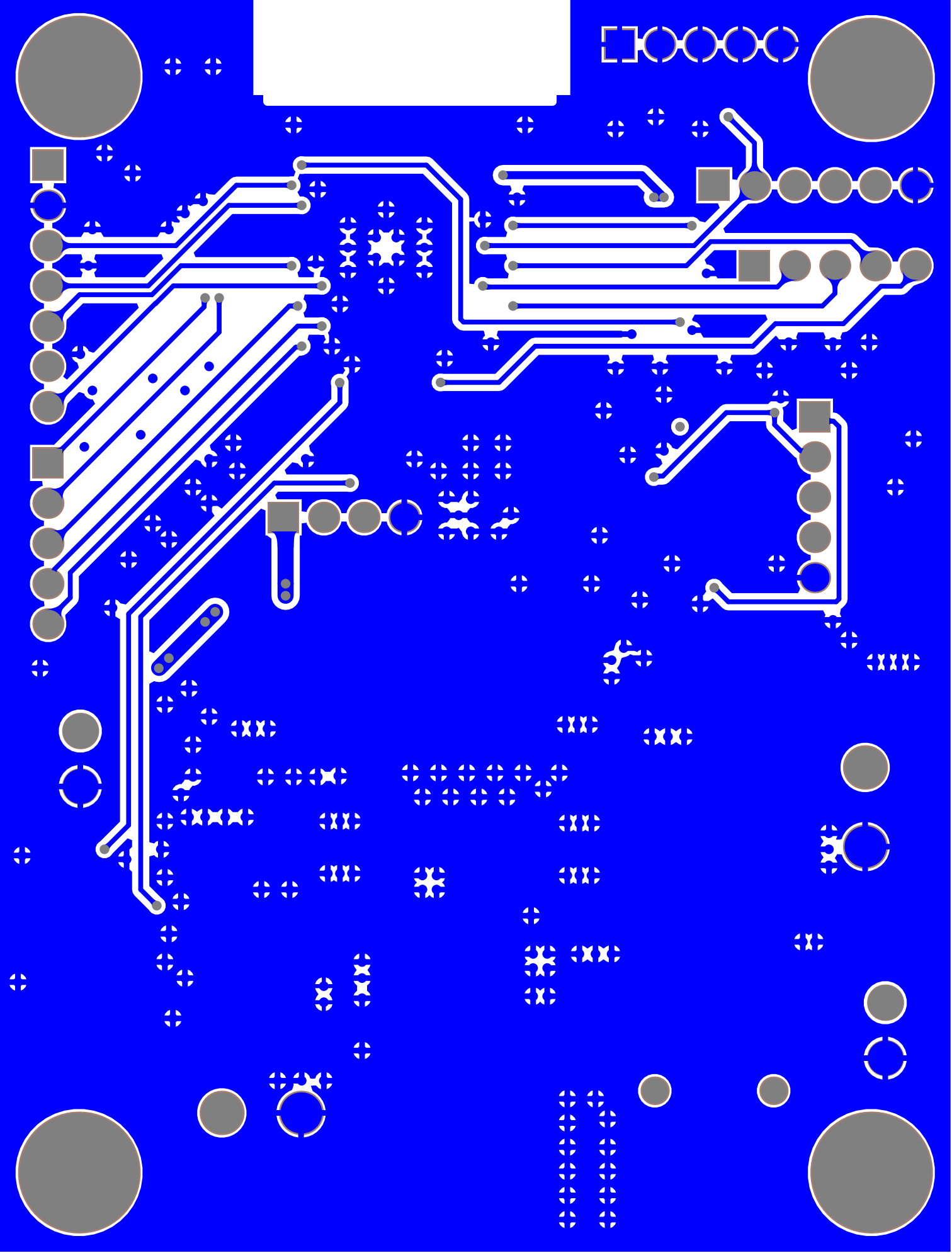


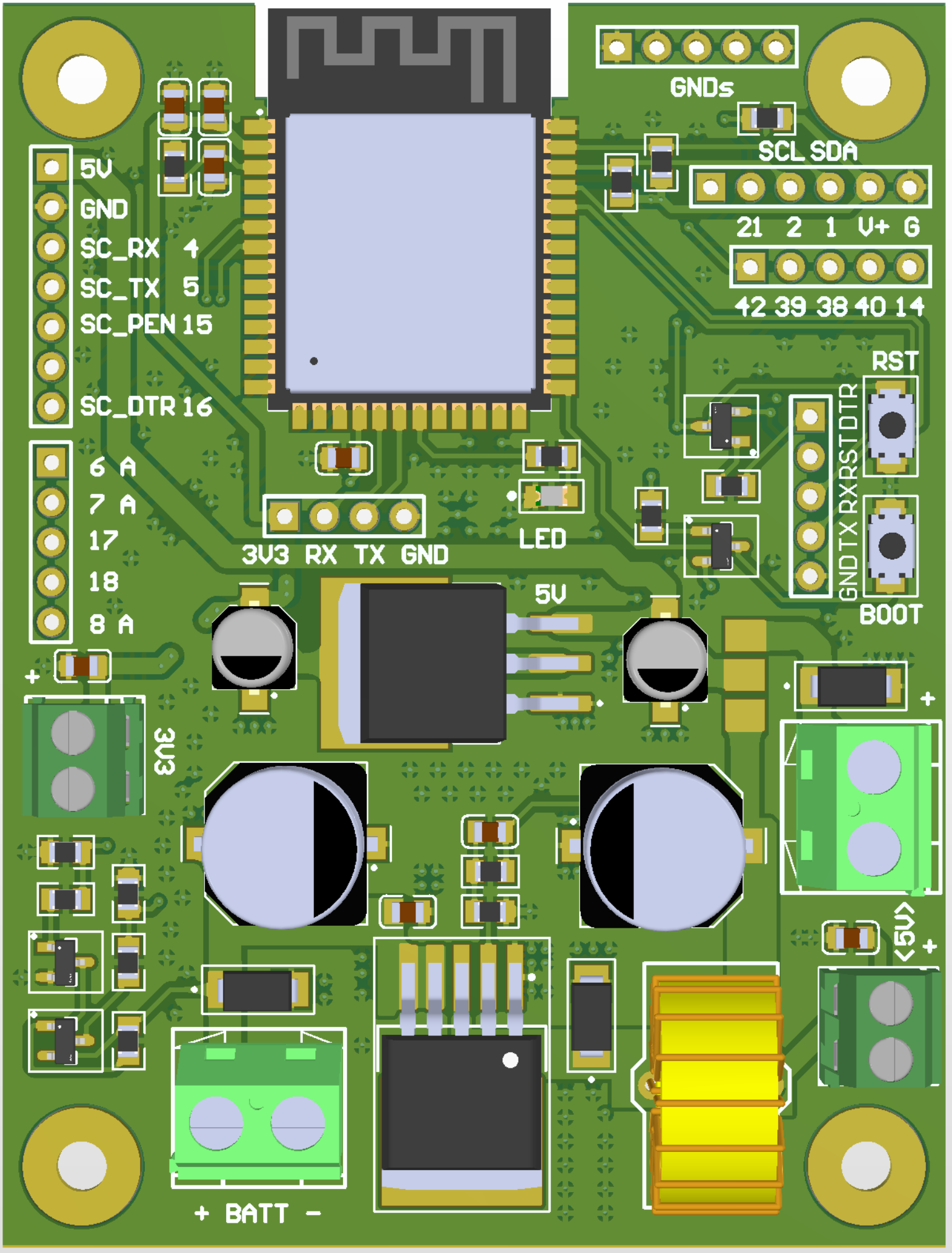
SERIAL SIGNALS HANDLING

DESIGN OF A PRINTED CIRCUIT BOARD FOR A BATTERY-POWERED WIRELESS SENSOR NETWORK

Title: DESIGN OF A PRINTED CIRCUIT BOARD FOR A BATTERY-POWERED WIRELESS SENSOR NETWORK		
Size: A3	Number:	Revision:
Date: 3/04/2026		Sheet of
File: C:\Users\..\ESP32_EC.SchDoc		Drawn By: PHAM NGUYEN MINH DUC


ANTENNA
GNDs
SCL SDA
21 2 1 V+ G
42 39 38 40 14
RST
GND TX RX IO0
BOOT
5V
GND
SC_RX 4
SC_TX 5
SC_PEN 15
SC_DTR 16
6 A
7 A
17
18
8 A
3V3 RX TX GND
LED
5V
3V3
<5V>
+ BATT -




GNDs
SCL SDA
21 2 1 V+ G
42 39 38 40 14
RST
BOOT
GNDTX RXRSTDTR
5V
GND
SC_RX 4
SC_TX 5
SC_PEN 15
SC_DTR 16
6 A
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8 A
3V3 RX TX GND
LED
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3V3
+ BATT -
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